TUTALY

Platform Design & Architecture Document

Version 1.1 · Phase 1 MVP · Nigeria Market

CONFIDENTIAL

For Designer · For Developer

UI/UX Screens · User Flows · Backend Architecture · Database Schema · API Endpoints · Security

SECTION 1 — PLATFORM OVERVIEW

Tutaly is a Nigeria-first professional web platform that connects job seekers and employers, aggregates company and salary intelligence, enables a full-featured marketplace for both digital and physical work-related products and services, and fosters professional networking — all in one place.

1.1 Platform Summary

Attribute	Detail
Platform Type	Web Application (Mobile Responsive)
Phase 1 Market	Nigeria
Future Roadmap	Global Expansion
Primary Language	English
Currency	Nigerian Naira (NGN) — Flutterwave
User Base	Job Seekers, Employers, Sellers, Buyers, Admins

1.2 Core Modules

Module	Description	Key Users	
Authentication	Registration, login, email verification, password reset	All	
Jobs	Job posting, search, filtering, apply, save, report	Seeker, Employer, Admin	
Company Reviews	Submit, moderate, display anonymised company reviews	All, Admin	
Salary Reviews	Submit and view anonymised salary data by role/industry	All	
Connect	Professional networking — feed, posts, follow, DM, like, comment	Seeker, Employer	
Shop (Marketplace)	Full marketplace: digital products, physical goods, and services with Flutterwave escrow checkout	All, Seller, Buyer, Admin	
Advertising	Banner, featured job, and sponsored ad placements	Employer, Admin	
Admin Dashboard	Full platform moderation, analytics, legal pages, email broadcast	Admin, Super Admin	

1.3 User Types & Permissions

User Type	Access Level	Key Capabilities	
Guest	Read-only public pages	View jobs, reviews, salaries, shop. Submit reviews/salaries anonymously.	
Job Seeker	Authenticated — seeker role	Apply to jobs, save jobs, full Connect access, purchase from Shop, apply to sell.	
Employer	Authenticated — employer role	Post jobs, manage applicants, Company Profile, post to Connect feed, purchase from Shop.	
Seller	Authenticated — approved seller role	List products/services, manage orders, receive payouts, view earnings dashboard.	
Admin	Authenticated — admin role	Moderate all content, manage users, edit legal pages, manage ads, send newsletters.	
Super Admin	Authenticated — super_admin role	All Admin capabilities + view revenue, view analytics, manage commissions.	

1.4 Design Principles

- Mobile-first responsive layout — Nigeria has predominantly mobile internet users
- Professional and trustworthy tone — LinkedIn meets Glassdoor meets a work-focused marketplace for Nigeria
- Primary palette: Deep Navy #0D1B2A (primary) + Teal #1D9E75 (accent) + White backgrounds
- Typography: Bold sans-serif headings (Arial/Inter), 16px base body copy
- Data-heavy pages use card + table hybrid layouts

- Moderation states always visually distinct: Pending (amber), Approved (green), Rejected (red)
- All forms include a progress indicator, inline validation, and clear error states
- Consistent 8px spacing grid; generous whitespace; no visual clutter
- Accessibility: minimum AA contrast ratio, focus states on all interactive elements

SECTION 2 — UI/UX SCREEN DESCRIPTIONS

Every screen below is described with its layout zones, components, interactive states, and responsive behaviour.

PUBLIC PAGES — Visible to all users including guests

Screen 1 — Home Page

Purpose: Primary marketing and discovery page. Converts guests to registered users and routes job seekers and employers to the right flow.

Layout Zones (Top to Bottom)

- Sticky Top Navigation Bar — Logo (left) · Jobs · Reviews · Salaries · Shop · Community (center) · Sign In (ghost) + Post a Job (teal CTA) (right) · Mobile: hamburger slide-out drawer
- Hero Section — Deep navy background · Bold headline · Search bar (Keyword + Location + Role) · Dual CTA: 'Find a Job' + 'Post a Job' · Logged-in state: 'Welcome back [Name]'
- Platform Stats Bar — 4 animated count-up blocks: Active Jobs · Companies Reviewed · Salaries Shared · Community Members
- Featured Jobs Carousel — 3–4 swipeable job cards with company logo, title, location, salary, Featured/Urgent badge
- Browse by Industry Grid — 8–12 icon tiles (Technology, Finance, Healthcare, etc.) with hover lift effect
- Company Reviews Snapshot — 3 review excerpt cards with star rating and sector badge
- Salary Insights Teaser — Horizontal bar chart for 4–5 top roles. CTA: 'Explore Full Salary Data'
- Shop Highlights — 3 featured product/service cards with thumbnail, category, price, 'View' button
- Join Community CTA Banner — Full-width teal banner with 'Join for Free' button
- Footer — 4-column link grid + all legal page links + copyright + social icons

Screen 2 — Jobs Page

- Page Header — Breadcrumb + result count + top search bar + 'Show Filters' toggle
- Filter Sidebar — Collapsible groups: Industry / Role / Category / Employment Type / Salary Type / Qualification / Experience Level / Work Mode / Date Posted
- Job Results List — Sort bar + job cards + pagination
- Job Detail Panel — Company logo + title + badges + Apply/Save/Share/Report buttons + description + requirements + company overview card

Job Card States

State	Visual Treatment
Featured	Gold/amber left border + 'Featured' badge
Urgent	Red left border + 'Urgent' badge
Saved	Bookmark icon fills teal
Applied	Card dims + 'Applied' green badge
Expired	Card grays out + 'Expired' badge + apply disabled
Guest apply attempt	Redirect to Sign In with message

Screen 3 — Company Reviews Page

- Search & Header — Company search with autocomplete + 'Write a Review' button (open to all)
- Company Cards Grid — Logo/initials · name · sector · star rating · review count · 'View Reviews' link
- Company Detail Page — Header with rating breakdown bar chart (Work-Life / Pay / Management / Culture) + 'Would Recommend' percentage + review list
- Individual Review Card — Star rating · position · pros (green border) · cons (red border) · recommend badge · report flag · admin status badge
- Write a Review Form — Company · sector · position · rating sliders · pros/cons textareas · recommend toggle · nickname · CAPTCHA · privacy note

Screen 4 — Salary Reviews Page

- Filter Bar — Industry (required) + Role + Location + Company (optional)
- Summary Cards Row — Average salary · salary range (min–max) · data points count · most common location
- Salary Range Chart — Horizontal bar chart per role showing Min / Median / Max in teal
- Data Table — Role / Average Salary / Range / Location / Submissions / Year — sortable, paginated
- Submit a Salary Panel — Industry / Company / Role / Amount / Pay Period / Location / Year · full anonymity note · CAPTCHA

Screen 5 — Shop (Marketplace) [UPDATED — FULL SCOPE]

Purpose: A comprehensive work-related marketplace covering digital products, physical goods, and professional services. All transactions processed through Tutaly via Flutterwave with an escrow model to protect buyers and sellers.

5A — Shop Homepage & Discovery

- Shop Header — Page title 'Tutaly Shop' · search bar · top-level category tabs (see Section 8 for full category tree)
- Category Navigation — Expandable sidebar or horizontal scroll tabs for all top-level categories with subcategory drill-down
- Product/Service Cards — Thumbnail · category badge · title · seller name · star rating · price (or 'Request Quote') · listing type badge (Product / Service / Digital)
- Hover state — card lifts, quick-view button appears

5B — Listing Detail Page

- Full-width thumbnail/gallery · Title · Category + subcategory breadcrumb · Listing type badge
- Seller profile mini-card: avatar, name, ratings, response time
- Pricing display — shows 'Price per Unit' (with unit and minimum quantity) OR 'Request Quote' button if seller selected quote pricing
- 'Final price may vary based on scope' note shown if seller enabled it
- Add to Cart / Buy Now buttons (hidden if listing is 'Request Quote' — replaced by quote request form)
- Tabs: Description · Reviews · Seller Info · Delivery & Refund Policy
- For digital products — file format + file size info + preview section (if seller provides one)
- For physical goods — estimated delivery timeline + bulk order note
- For services — scope of work description + typical turnaround

5C — Pricing Model

Sellers choose ONE pricing type per listing when creating it:

Pricing Type	How It Works
Price per Unit	Seller sets a fixed price, unit (e.g. per item / per person / per hour), and minimum order quantity. Buyer adds to cart and pays directly.
Request a Quote	No price shown. Buyer clicks 'Request Quote', fills a brief requirement form. Seller responds with a custom price. Buyer accepts and pays through Tutaly.

- Optional flag: 'Final price may vary based on scope' — shown on listing for service-type items

5D — Escrow & Payment Flow

All payments are held by the platform until delivery is confirmed. This protects buyers and ensures sellers are paid fairly.

Step	Action
1. Payment	Buyer pays in full via Flutterwave. Funds held in escrow by Tutaly.
2. Delivery	Seller delivers the product or service.
3. Confirmation window	Buyer has 3–6 days after delivery to confirm receipt OR report an issue.
4a. Buyer confirms	Payment released to seller immediately (minus platform commission).
4b. No response	System automatically releases payment to seller after the window expires.
4c. Buyer reports issue	Payment frozen. Admin reviews and decides: refund buyer or release to seller.
5. Non-delivery	If seller fails to deliver within agreed timeframe, buyer receives full refund.

5E — Commission Structure

- Tutaly automatically deducts 20% commission from every completed transaction
- Seller receives 80% of the order value after confirmation
- Commission tracked per order in the database
- Super Admin views gross revenue, total commission earned, and seller payables in Revenue tab

5F — Shopping Cart & Checkout

- Cart — Slide-out sidebar: thumbnail + title + price + remove button + subtotal + 'Proceed to Checkout'
- Checkout — Order summary (right) + billing details form (name, email, phone) + Flutterwave payment widget + Place Order button
- For 'Request a Quote' items — separate quote request flow (not cart-based)

5G — Post-Purchase Experience

- Order Confirmation Page — Success icon + order reference + product name + Download button (digital) or delivery info (physical/service)
- My Purchases Page (Buyer) — Table: Product / Date / Amount / Download or Track / Rate Product button
- Digital downloads — signed S3 URL generated per request (1 hour expiry), re-downloadable unlimited times
- Physical orders — status updates: Pending → Processing → Dispatched → Delivered → Confirmed

5H — Seller Dashboard

- Nav: My Listings · Add Listing · Orders · Earnings · Settings
- Stats cards: Total Sales · Total Earnings (after 20%) · Active Listings · Pending Orders

- Add Listing form: Title · Description · Category + Subcategory · Listing Type (Digital / Physical / Service) · Pricing Type (Per Unit or Quote) · Price + Unit + Min Quantity (if Per Unit) · Thumbnail · File upload (digital) or delivery info (physical) · 'Final price may vary' toggle
- Orders tab — table with buyer name, product, date, amount, status, and action buttons (Mark Delivered, View, Message Buyer)
- Earnings tab — transaction history with commission breakdown, payout status

5l — Request a Quote Flow

- Buyer clicks 'Request Quote' on listing
- Form opens: requirement description + budget range (optional) + deadline (optional) + contact preference
- Seller receives notification and responds with custom quote via platform messaging
- Buyer accepts quote → system generates checkout link → buyer pays → escrow flow begins

Screen 6 — Connect (Networking)

Desktop 3-column layout, mobile single column.

- Left Sidebar — Avatar + name + @username + follower/following count + navigation (Feed / My Posts / Saved / Discover / Messages / Notifications)
- Center Feed — Create Post box + reverse-chronological post cards (text + optional image) with Like / Comment / Share / Report
- Right Sidebar — 'People You May Know' + trending topics (desktop only)
- Profile Page — Cover photo + avatar + name + bio + Follow/Unfollow + Message + stats row + posts grid
- Messages (DM) — Conversation list (left) + chat window with speech bubbles (right)
- Notifications Page — Activity list with mark-all-read

AUTH PAGES

Screen 7 — Sign In Page

- Centered card on white/light background · Logo at top
- Email + Password (show/hide toggle) · 'Forgot Password?' link
- Sign In button (full width, teal) · 'or' divider · 'Sign Up' link
- Error states: inline under field · Loading: spinner on button

Screen 8 — Sign Up Flow (3 Steps)

- Step 1 — Choose Role: Two large cards (Job Seeker / Employer). Selected: teal border + checkmark. Progress bar top.
- Step 2a (Seeker) — Full Name / Email / Password (strength meter) / Confirm Password / Date of Birth (18+ validated) / Terms checkbox / CAPTCHA
- Step 2b (Employer) — Company Name / Full Name / Work Email / Password / Company Size dropdown / Terms checkbox / CAPTCHA
- Step 3 — Email Verification: envelope illustration + 'Check Your Inbox' heading + resend link (60s cooldown)

DASHBOARD PAGES — Authenticated Users

Screen 9 — Job Seeker Dashboard

- Sidebar nav: Overview / Saved Jobs / My Applications / Profile / Account Settings
- Overview Tab — Welcome message + stats cards (Applications / Saved / Profile Views) + recommended jobs + recent applications table
- My Applications Tab — Full list with visual status pipeline: Applied → Under Review → Shortlisted → Offer / Rejected
- Profile Tab — Photo / name / headline / bio / skills / location / experience / education / CV upload

Screen 10 — Employer Dashboard

- Sidebar nav: Overview / Post a Job / My Jobs / Applicants / Company Profile / Settings
- Overview Tab — Stats (Active Jobs / Total Applicants / Job Views / Applications this week) + recent applicants table
- Post a Job Tab — Multi-section form: Job Details / Requirements / Application Settings + Submit for Review button
- My Jobs Tab — Cards with status badges (Pending/Active/Expired/Removed) + Edit/Boost/Pause/Delete actions
- Applicants Tab — Dropdown to select job + applicants table with CV download + status update dropdown
- Company Profile Tab — Logo / about / sector / website / social links

Screen 11 — Admin Dashboard

- Sidebar: Overview / Users / Jobs / Reviews / Salaries / Shop & Sellers / Ads / Revenue / Legal Pages / Email & Newsletters / Settings
- Overview Tab — KPI cards + user signup line chart + quick action buttons
- Users Tab — Searchable table with View / Suspend / Activate / Delete row actions + bulk actions
- Jobs Tab — Pending/Active/Expired/Removed filter + Approve/Remove/View actions

- Reviews Tab — Moderation queue: Approve / Reject / Edit per review
- Shop & Sellers Tab — Seller applications queue + products table + commission tracker + order dispute queue
- Revenue Tab (Super Admin) — Gross revenue / commissions / seller payables + full transaction table
- Legal Pages Tab — 12 pages with WYSIWYG editor per page
- Email & Newsletters Tab — Compose + audience select + send/schedule + send history

Screen 12 — Free Job Posting Landing Page

- Marketing page: free vs premium comparison table · inline quick-post form · CTA: 'Create Employer Account'

Screen 13 — Settings Page

- Tabs: Account (change email/password/delete account) / Notifications (per-category toggles) / Privacy (visibility + follow approval) / Cookies

Screen 14 — Legal Pages (×12)

- Consistent template: Logo + page title + last updated date + table of contents + H2/H3 sections + footer with links to all other legal pages
- Pages: Terms of Service / Privacy Policy / Disclaimer / Community Guidelines / Review Policy / Marketplace Policy / Refund Policy / Advertiser Policy / Employer Policy / Cookie Policy / About Us / Contact Us

SECTION 3 — USER FLOW DIAGRAMS

Flow 1 — Job Seeker Registration to First Application

Step	Screen / Action	System Action
1	Arrives on Home Page	Serve public home page
2	Clicks 'Sign Up'	Redirect to Sign Up page
3	Selects 'Job Seeker'	Sets role = seeker in form state
4	Fills form + CAPTCHA	Client validates: age 18+, password strength, email format
5	Submits form	POST /auth/register → creates user (unverified) + sends verification email
6	Email verification screen	Awaits email link click
7	Clicks link in email	GET /auth/verify-email?token=X → email_verified = true
8	Redirected to dashboard	JWT issued, session started
9	Completes profile	Profile completeness indicator shown
10	Navigates to Jobs page	GET /jobs with query params
11	Clicks job card	GET /jobs/:id
12	Clicks 'Apply Now'	POST /jobs/:id/apply → creates application record
13	Application in dashboard	Status: Applied

Flow 2 — Employer Posts a Job

Step	Screen / Action	System Action
1	Employer registers	POST /auth/register (role = employer)
2	Verifies email	Email verification flow
3	Completes Company Profile	PUT /profiles/employer/:id
4	Fills job post form	Post a Job form rendered
5	Submits job	POST /jobs → status: pending_review

6	Pending badge shown	Awaiting admin approval
7	Admin approves	PATCH /admin/jobs/:id → status: active
8	Job goes live	Employer notified by email
9	Seekers apply	POST /jobs/:id/apply creates applications
10	Employer views applicants	GET /jobs/:id/applications
11	Employer updates status	PATCH /applications/:id → status: shortlisted

Flow 3 — Guest Submits Company Review

Step	Screen / Action	System Action
1	Guest visits Reviews page	GET /reviews/companies
2	Clicks 'Write a Review'	No auth required
3	Review form opens	user_id = null in submission
4	Fills fields + CAPTCHA	Client validation
5	Submits form	POST /reviews → status: pending + hashed IP+UA stored
6	Success message shown	Admin notified
7	Admin approves/rejects	PATCH /admin/reviews/:id
8	If approved	Review published with nickname only, real name never shown

Flow 4 — Shop: Product Purchase & Download (Digital)

Step	Screen / Action	System Action
1	User browses Shop	GET /shop/products
2	Opens product detail	GET /shop/products/:id
3	Adds to cart	POST /shop/cart
4	Proceeds to checkout	GET /shop/cart
5	Fills billing details	Client-side form validation

6	Clicks 'Pay Now'	POST /shop/checkout → pending order + Flutterwave initialised
7	Flutterwave payment done	Webhook POST /shop/webhook/flutterwave
8	HMAC + idempotency check	If valid: order status → paid
9	Commission deducted (20%)	commission_amount = total × 0.20 stored
10	Confirmation + Download	Signed S3 URL generated (1 hour expiry)
11	Re-download anytime	New signed URL generated per request

Flow 5 — Shop: Physical/Service Order with Escrow

Step	Screen / Action	System Action
1	Buyer places order	POST /shop/checkout → escrow order created
2	Payment held by platform	Flutterwave webhook confirms payment; status = paid_escrow
3	Seller delivers	Seller marks order as 'Delivered' in dashboard
4	Buyer confirmation window opens	System starts 3–6 day timer
5a	Buyer confirms delivery	POST /shop/orders/:id/confirm → status = complete → payment released to seller
5b	No response within window	Cron job: auto-release payment → status = auto_complete
5c	Buyer reports issue	POST /shop/orders/:id/dispute → status = disputed → admin notified
6	Admin reviews dispute	PATCH /admin/disputes/:id → refund or release
7	Non-delivery	If deadline passes with no delivery → auto-refund buyer

Flow 6 — Request a Quote

Step	Screen / Action	System Action
1	Buyer clicks 'Request Quote'	Quote request form opens on listing page

2	Fills requirements + budget	POST /shop/quotes → quote request saved, seller notified
3	Seller receives notification	GET /shop/my-quotes → seller views request
4	Seller sends custom quote	POST /shop/quotes/:id/respond → buyer notified
5	Buyer accepts quote	POST /shop/quotes/:id/accept → checkout link generated
6	Buyer pays	Standard Flutterwave checkout → escrow flow begins (Flow 5)

Flow 7 — Seller Onboarding & Listing Upload

Step	Screen / Action	System Action
1	User applies to become seller	POST /shop/seller/apply
2	Fills seller application	Status: pending — admin notified
3	Admin approves	PATCH /admin/sellers/:id → seller_status: approved
4	User receives approval email	Seller Dashboard unlocked
5	Seller creates listing	POST /shop/products → file/details stored, status: active
6	Buyer purchases	Flow 4 or 5 depending on listing type
7	Seller views earnings	GET /shop/my-sales → earnings after 20% commission

Flow 8 — Admin Moderation Loop

Step	Item Type	Admin Action	Outcome
1	New Job Post	Review + Approve or Remove	Active on Jobs page or removed with employer notification
2	Company Review	Check for abuse + Approve or Reject	Published anonymously or discarded
3	Seller Application	Review details + Approve or Reject	Seller Dashboard unlocked or rejection email
4	Order Dispute	Review evidence + Refund buyer or Release to seller	Order resolved, parties notified

5	Reported Content	View report + Remove or Dismiss	Content removed or report dismissed
6	User Report	View flagged user + Suspend or Delete	User notified, access revoked

Flow 9 — Connect / Networking

Step	Action	System Action
1	Visits Discover People	GET /connect/discover
2	Clicks Follow	POST /connect/follow/:userId → status: pending
3	Target receives notification	Notification: '[Name] wants to follow you'
4	Target accepts/rejects	PATCH /connect/follow/:reqId → accepted or rejected
5	Feed updated	Feed fan-out job queued (Bull)
6	Creates a post	POST /connect/posts
7	Post fans out to followers	Background job updates feed cache in Redis
8	Users like/comment	POST /connect/posts/:id/like or /comments
9	Sends direct message	POST /connect/messages/:userId

Flow 10 — Password Reset

Step	Action	System Action
1	Clicks 'Forgot Password?'	Render Forgot Password page
2	Enters email, submits	POST /auth/forgot-password → generate reset token (1 hour expiry) → send email
3	Success message shown	'If this email exists, a reset link has been sent.'
4	Clicks link in email	GET /auth/reset-password?token=X → validate token
5	New Password form shown	Client: password strength validation
6	Submits new password	POST /auth/reset-password → bcrypt hash → mark token used → invalidate all refresh tokens

7	Redirected to Sign In	Password updated toast shown

SECTION 4 — BACKEND ARCHITECTURE

4.1 Technology Stack

Layer	Technology / Justification
Frontend	Next.js 14 (App Router) + Tailwind CSS — SSR for SEO on Jobs/Reviews/Salaries/Shop pages
Backend API	Node.js + NestJS (REST) — TypeScript, structured, scalable for this module count
Primary Database	PostgreSQL — relational data, ACID transactions, full-text search
Cache & Sessions	Redis — JWT refresh tokens, feed cache, rate limiting, Bull job queues
File Storage	AWS S3 (private bucket) — secure digital product files; signed URL downloads
Authentication	JWT (access 15min + refresh 7d) + bcrypt 12 rounds
Email Service	SendGrid or Mailgun — transactional emails + newsletter broadcasts
Payments	Flutterwave — NGN support; inline checkout; webhook + escrow management
Background Jobs	Bull (Redis-backed queue) — email sends, feed fan-out, escrow timers, dispute checks
CDN	Cloudflare — DDoS protection, static asset caching, SSL termination
Search	PostgreSQL Full-Text Search (Phase 1) — upgrade to Elasticsearch in Phase 2
Monitoring	Sentry + UptimeRobot — error tracking, performance monitoring
Hosting	Vercel (Frontend) + AWS EC2 or Railway (API)
CI/CD	GitHub Actions + Docker — automated test and deploy pipeline

4.2 Service Architecture

Service	Responsibilities / External Dependencies
Auth Service	Registration, login, JWT, email verification, password reset, account deletion — SendGrid, Redis
User Service	Profile management (seeker + employer), avatar/CV upload, seller application — AWS S3
Job Service	Job CRUD, search/filter, applications, saved jobs, report job — PostgreSQL full-text index
Review Service	Company review submission, moderation, approval, company aggregates — internal

Salary Service	Salary data submission, aggregation queries (avg/min/max per role/industry) — internal
Connect Service	Posts, feed, likes, comments, follows, DMs, notifications, discover — Redis, Bull
Shop Service	Product/service CRUD, cart, checkout, escrow, quote flow, Flutterwave webhook, downloads, seller earnings, ratings — Flutterwave, AWS S3
Escrow Service	Payment holds, delivery confirmation, dispute handling, auto-release cron jobs — Flutterwave, Bull
Admin Service	User management, moderation queue, legal pages, newsletter, ads, revenue dashboard, dispute resolution — SendGrid
Notification Service	In-app + email notifications triggered by system events — SendGrid, Redis pub/sub

4.3 Architecture Layers

Layer	Description
Client Layer	Next.js web app on Vercel. Server Components for public pages (SEO), Client Components for interactive dashboards. HTTPS only.
CDN / Proxy Layer	Cloudflare handles SSL termination, DDoS mitigation, static asset caching, edge rate limiting.
Load Balancer	Nginx routes requests to API containers. Handles SSL offloading and horizontal scaling.
API Application Layer	NestJS REST API in Docker containers. Stateless — all session state in Redis.
Data Layer	PostgreSQL (primary DB + read replica) + Redis (cache, queues, sessions) + AWS S3 (file storage).
Background Job Layer	Bull queue workers: email dispatch, feed fan-out, escrow auto-release timers, dispute deadline checks.
External Services	Flutterwave (payments + escrow), SendGrid (email), Google reCAPTCHA (spam), AWS S3 (files), Sentry (errors).

4.4 Feed Architecture

Fan-out on write strategy for performance: post created → Bull job queued → worker fetches all followers → adds post_id to each follower's Redis sorted set (scored by timestamp) → feed load reads Redis → fetches post details from PostgreSQL. Feed TTL in Redis: 7 days. Older posts served directly from PostgreSQL.

4.5 Payment & Escrow Architecture

- Buyer initiates checkout → POST /shop/checkout → order created with status: pending_payment
- Flutterwave inline checkout initialised client-side with public key
- Buyer completes payment on Flutterwave
- Flutterwave sends POST webhook to /shop/webhook/flutterwave
- Server verifies HMAC-SHA512 signature — rejects if invalid
- Idempotency check: if order already processed, return 200 (prevent double-processing)
- If valid: order status → paid_escrow · commission_amount = total × 0.20 stored · seller notified · escrow timer started
- Bull cron job monitors escrow timer. On expiry with no buyer action: auto-releases payment, status → auto_complete
- Dispute raised: status → disputed · payment frozen · admin notified · admin resolves via /admin/disputes/:id

CRITICAL: Never trust client-side payment confirmation. All order status updates happen only via the verified webhook.

SECTION 5 — DATABASE SCHEMA

PostgreSQL. All tables use UUID primary keys. Every table includes created_at and updated_at TIMESTAMP columns auto-managed. Relationships use foreign keys with appropriate cascade rules.

5.1 users

Column	Type / Notes
id	UUID PRIMARY KEY, gen_random_uuid()
email	VARCHAR(255) UNIQUE NOT NULL — indexed
password_hash	VARCHAR(255) NOT NULL — bcrypt 12 rounds
role	ENUM: seeker | employer | admin | super_admin — NOT NULL
status	ENUM: active | suspended | deleted — default: active
email_verified	BOOLEAN default false
date_of_birth	DATE NOT NULL — must be 18+
tos_agreed_at	TIMESTAMP NOT NULL — logged at registration
username	VARCHAR(50) UNIQUE NOT NULL — format: name_slug_XXXX
full_name	VARCHAR(255) NOT NULL
avatar_url	VARCHAR(500) nullable — S3 path

5.2 seeker_profiles

Column	Type / Notes
user_id	UUID FK → users.id ON DELETE CASCADE PRIMARY KEY
headline	VARCHAR(255) nullable
bio	TEXT nullable
skills	TEXT[] PostgreSQL array
location	VARCHAR(255) nullable
cv_url	VARCHAR(500) nullable — private S3 path
profile_visibility	ENUM: public | connections — default: public
follow_approval	ENUM: auto | manual — default: auto
seller_status	ENUM: none | pending | approved | suspended — default: none

5.3 employer_profiles

Column	Type / Notes
user_id	UUID FK → users.id ON DELETE CASCADE PRIMARY KEY
company_name	VARCHAR(255) NOT NULL
about	TEXT nullable
sector	VARCHAR(255) nullable
company_size	ENUM: 1-10 | 11-50 | 51-200 | 201-500 | 500+
website_url	VARCHAR(500) nullable
logo_url	VARCHAR(500) nullable — S3 path
social_links	JSONB nullable — { linkedin, twitter, etc. }

5.4 jobs

Column	Type / Notes
id	UUID PRIMARY KEY
employer_id	UUID FK → users.id NOT NULL — indexed
title	VARCHAR(255) NOT NULL — full-text indexed
description	TEXT NOT NULL — full-text indexed
location	VARCHAR(255) NOT NULL
is_remote	BOOLEAN default false
work_mode	ENUM: remote | hybrid | onsite
employment_type	ENUM: full_time | part_time | contract | internship
salary_min / salary_max	NUMERIC(12,2) nullable
salary_type	ENUM: hourly | monthly | annual — nullable
industry	VARCHAR(255) NOT NULL — indexed
category / role	VARCHAR(255) indexed
qualification	ENUM: none | ond | hnd | bsc | msc | phd
experience_level	ENUM: entry | mid | senior | executive
apply_method	ENUM: external_url | in_platform
apply_url	VARCHAR(500) nullable — required if external_url
deadline	TIMESTAMP nullable

status	ENUM: pending_review | active | expired | removed — default: pending_review
is_featured / is_urgent	BOOLEAN default false

5.5 applications + saved_jobs

Column	Type / Notes
— applications —	
id	UUID PRIMARY KEY
job_id / seeker_id	UUID FKs — NOT NULL
status	ENUM: applied | reviewing | shortlisted | rejected | offered — default: applied
cover_note	TEXT nullable
applied_at	TIMESTAMP NOT NULL, default NOW()
UNIQUE	(job_id, seeker_id) — prevents duplicate applications
— saved_jobs —	
id	UUID PRIMARY KEY
job_id / seeker_id	UUID FKs — NOT NULL
saved_at	TIMESTAMP NOT NULL, default NOW()
UNIQUE	(job_id, seeker_id)

5.6 company_reviews

Column	Type / Notes
id	UUID PRIMARY KEY
company_name	VARCHAR(255) NOT NULL — indexed
sector / position / department	VARCHAR(255) nullable
rating_overall	SMALLINT 1–5 NOT NULL
rating_work_life / rating_pay / rating_management / rating_culture	SMALLINT 1–5 nullable
pros	TEXT NOT NULL
cons	TEXT NOT NULL

recommend	BOOLEAN NOT NULL
display_name	VARCHAR(100) NOT NULL — nickname, never real name
submitter_hash	VARCHAR(255) internal only — hashed IP+UA, never exposed in API
user_id	UUID nullable FK → users.id (null for guest submissions)
status	ENUM: pending | approved | rejected — default: pending

5.7 salary_reviews

Column	Type / Notes
id	UUID PRIMARY KEY
industry	VARCHAR(255) NOT NULL — indexed
company	VARCHAR(255) nullable
role	VARCHAR(255) NOT NULL — indexed
salary_amount	NUMERIC(12,2) NOT NULL
currency	CHAR(3) default 'NGN'
salary_period	ENUM: monthly | annual
location	VARCHAR(255) nullable
submission_year	SMALLINT NOT NULL
No user_id stored	Fully anonymous — no linkage to any user account possible

5.8 shop_products

Column	Type / Notes
id	UUID PRIMARY KEY
seller_id	UUID FK → users.id NOT NULL (must be approved seller)
title	VARCHAR(255) NOT NULL
description	TEXT NOT NULL
listing_type	ENUM: digital | physical | service — NOT NULL
category	VARCHAR(255) NOT NULL — top-level category
subcategory	VARCHAR(255) NOT NULL — subcategory from category tree
pricing_type	ENUM: per_unit | request_quote — NOT NULL
price	NUMERIC(10,2) nullable — required if pricing_type = per_unit
price_unit	VARCHAR(100) nullable — e.g. 'per item', 'per hour', 'per person'

min_quantity	INT default 1 — required if pricing_type = per_unit
price_may_vary	BOOLEAN default false — 'Final price may vary based on scope'
file_s3_key	VARCHAR(500) nullable — private S3 key for digital products, never exposed directly
thumbnail_url	VARCHAR(500) nullable
status	ENUM: active | removed — default: active
is_work_related_confirmed	BOOLEAN NOT NULL — required for Other/Uncategorized listings

5.9 orders

Column	Type / Notes
id	UUID PRIMARY KEY
buyer_id	UUID FK → users.id NOT NULL
product_id	UUID FK → shop_products.id NOT NULL
seller_id	UUID FK → users.id NOT NULL — denormalised for query performance
amount_paid	NUMERIC(10,2) NOT NULL
commission_amount	NUMERIC(10,2) NOT NULL — amount_paid × 0.20
seller_earnings	NUMERIC(10,2) NOT NULL — amount_paid × 0.80
payment_ref	VARCHAR(255) UNIQUE — Flutterwave transaction reference
status	ENUM: pending_payment | paid_escrow | delivered | complete | auto_complete | disputed | refunded — default: pending_payment
escrow_release_at	TIMESTAMP nullable — calculated on delivery confirmation: delivery_at + 3–6 days
delivery_confirmed_at	TIMESTAMP nullable — when buyer confirmed
delivered_at	TIMESTAMP nullable — when seller marked delivered
download_count	INT default 0 — for digital products

5.10 quote_requests

Column	Type / Notes
id	UUID PRIMARY KEY
product_id	UUID FK → shop_products.id NOT NULL
buyer_id	UUID FK → users.id NOT NULL
seller_id	UUID FK → users.id NOT NULL

requirements	TEXT NOT NULL — buyer's scope description
budget_range	VARCHAR(100) nullable
deadline_requested	DATE nullable
status	ENUM: pending | quoted | accepted | rejected | expired — default: pending
quoted_price	NUMERIC(10,2) nullable — seller's response
seller_notes	TEXT nullable — seller's response notes
checkout_link_generated	BOOLEAN default false
expires_at	TIMESTAMP — 7 days from creation if no seller response

5.11 order_disputes

Column	Type / Notes
id	UUID PRIMARY KEY
order_id	UUID FK → orders.id NOT NULL UNIQUE
raised_by	UUID FK → users.id NOT NULL — the buyer
reason	TEXT NOT NULL
evidence_urls	TEXT[] nullable — S3 paths to uploaded evidence
status	ENUM: open | resolved_refund | resolved_release — default: open
resolved_by	UUID nullable FK → users.id — admin who resolved
resolution_notes	TEXT nullable
resolved_at	TIMESTAMP nullable

5.12 Connect: posts, follows, messages

Table / Column	Type / Notes
— posts —	id UUID · author_id FK · content TEXT NOT NULL · image_url nullable · likes_count INT (denormalised) · comments_count INT (denormalised)
— post_likes —	UNIQUE constraint on (post_id, user_id)
— post_comments —	post_id FK · author_id FK · body TEXT NOT NULL
— follows —	follower_id FK · followee_id FK · status ENUM: pending | accepted | rejected · UNIQUE (follower_id, followee_id)
— messages —	sender_id FK · receiver_id FK · body TEXT NOT NULL · read_at TIMESTAMP nullable · composite index on (sender_id, receiver_id)

— reports —	reporter_id FK · target_type ENUM: post | review | job | user · target_id UUID · reason TEXT NOT NULL

5.13 Supporting Tables

Table	Key Columns / Purpose
email_verifications	user_id FK · token VARCHAR UNIQUE · expires_at · used_at nullable — expire after 24 hours
password_resets	user_id FK · token VARCHAR UNIQUE · expires_at · used_at nullable — expire after 1 hour
notifications	user_id FK · type VARCHAR · message TEXT · is_read BOOLEAN · link VARCHAR nullable
legal_pages	slug VARCHAR UNIQUE · title VARCHAR · content TEXT · updated_by FK · updated_at
ads	advertiser_id FK · type ENUM: banner | featured_job | sponsored · image_url · target_url · placement · starts_at · ends_at · status
newsletter_sends	subject · body · audience ENUM: all | seekers | employers · sent_by FK · sent_at · recipient_count
product_ratings	product_id FK · buyer_id FK · rating SMALLINT 1-5 · comment nullable — UNIQUE (product_id, buyer_id) — verified buyers only
seller_applications	user_id FK · bio TEXT · category_focus VARCHAR · status ENUM: pending | approved | rejected · reviewed_by FK nullable

SECTION 6 — API ENDPOINT REFERENCE

Base URL: /api/v1. All authenticated endpoints require Authorization: Bearer <access_token>. Role guards enforced server-side via NestJS guards.

Success: { success: true, data: {}, meta: { page, total } } | Error: { success: false, message: string, errors: [] }

6.1 Auth Endpoints

Method + Endpoint	Auth	Description
POST /auth/register	None	Register new user. Validates age 18+. Sends verification email.
POST /auth/verify-email	None	Verify email. Returns JWT tokens.
POST /auth/login	None	Login. Returns access_token (15min) + sets refresh_token httpOnly cookie (7d).
POST /auth/refresh	Cookie	Issue new access token. Rotates refresh token.
POST /auth/forgot-password	None	Send reset email. Always returns 200 (no user enumeration).
POST /auth/reset-password	None	Reset password. Invalidates all refresh tokens.
GET /auth/me	Any Auth	Return current user profile.
PUT /auth/change-password	Any Auth	Change password.
DELETE /auth/account	Any Auth	Soft-delete + anonymise account.
POST /auth/logout	Any Auth	Invalidate refresh token.

6.2 Job Endpoints

Method + Endpoint	Auth	Description
GET /jobs	None	List/search jobs with all filter query params.
GET /jobs/:id	None	Get job detail. Increments view_count.
POST /jobs	Employer	Create job post. Status: pending_review.
PUT /jobs/:id	Employer (own)	Update job.
DELETE /jobs/:id	Employer / Admin	Soft-delete job.
POST /jobs/:id/apply	Seeker	Apply. Body: { cover_note? }. Fails if duplicate.

POST /jobs/:id/save	Seeker	Save job to bookmarks.
DELETE /jobs/:id/save	Seeker	Remove saved job.
POST /jobs/:id/report	Any Auth	Report a job.
PATCH /jobs/:id/status	Admin	Update job status.
GET /jobs/my-jobs	Employer	List own job posts.
GET /jobs/saved	Seeker	List saved jobs.
GET /jobs/applications	Seeker	List seeker's own applications.
GET /jobs/:id/applications	Employer (own)	List applicants for a job.
PATCH /jobs/applications/:id	Employer (own)	Update application status.

6.3 Reviews & Salary Endpoints

Method + Endpoint	Auth	Description
GET /reviews/companies	None	List companies with review data.
GET /reviews/companies/:name	None	Company profile with ratings + reviews.
POST /reviews	None (public)	Submit review. user_id added if authenticated, null if guest.
POST /reviews/:id/report	Any Auth	Report a review.
PATCH /reviews/:id	Admin	Update review status.
GET /salaries	None	Get aggregated salary data.
POST /salaries	None (public)	Submit salary data. No user_id stored.

6.4 Shop Endpoints — Products & Listings

Method + Endpoint	Auth	Description
GET /shop/products	None	List products/services. Query: category, subcategory, listing_type, pricing_type, search, sort, page.
GET /shop/products/:id	None	Get listing detail with seller info and ratings.

POST /shop/products	Approved Seller	Create listing. Multipart form with optional digital file + thumbnail.
PUT /shop/products/:id	Seller (own)	Update listing details.
DELETE /shop/products/:id	Seller (own) / Admin	Remove listing.
POST /shop/products/:id/rate	Buyer (verified)	Submit product rating. One per buyer per product.

6.5 Shop Endpoints — Cart & Checkout

Method + Endpoint	Auth	Description
GET /shop/cart	Any Auth	Get current cart.
POST /shop/cart	Any Auth	Add item to cart.
DELETE /shop/cart/:product_id	Any Auth	Remove item from cart.
POST /shop/checkout	Any Auth	Create pending order + initialise Flutterwave. Returns payment link/data.
POST /shop/webhook/flutterwave	HMAC	Receive + process payment webhook. Verifies HMAC signature. Updates order status.
GET /shop/orders/:id/download	Buyer (paid)	Generate signed S3 URL (1 hour expiry). Increments download_count.
GET /shop/my-purchases	Any Auth	List all paid orders with download/tracking access.
POST /shop/orders/:id/confirm	Buyer (own order)	Buyer confirms delivery. Triggers payment release.
POST /shop/orders/:id/dispute	Buyer (own order)	Raise dispute. Body: { reason, evidence_urls? }. Freezes escrow.
PATCH /shop/orders/:id/delivered	Seller (own order)	Seller marks order as delivered. Starts confirmation window.
GET /shop/my-sales	Approved Seller	List sales with commission breakdown.

6.6 Shop Endpoints — Quotes

Method + Endpoint	Auth	Description
POST /shop/quotes	Any Auth	Buyer submits quote request. Body: { product_id, requirements, budget_range?, deadline? }.

GET /shop/quotes	Any Auth	List own quote requests (buyer) or received quotes (seller).
POST /shop/quotes/:id/respond	Seller (own listing)	Seller responds with quoted price + notes.
POST /shop/quotes/:id/accept	Buyer (own quote)	Buyer accepts quote. Generates checkout link.
POST /shop/quotes/:id/reject	Buyer (own quote)	Buyer rejects quote.

6.7 Shop Endpoints — Seller Onboarding

Method + Endpoint	Auth	Description
POST /shop/seller/apply	Any Auth	Apply to become a seller. Creates pending seller application.
GET /shop/seller/status	Any Auth	Check current seller application status.

6.8 Connect Endpoints

Method + Endpoint	Auth	Description
GET /connect/feed	Seeker / Employer	Get paginated feed. Reads from Redis, falls back to PostgreSQL.
POST /connect/posts	Any Auth	Create post.
DELETE /connect/posts/:id	Own / Admin	Delete post.
POST /connect/posts/:id/like	Any Auth	Like a post.
DELETE /connect/posts/:id/like	Any Auth	Unlike a post.
GET /connect/posts/:id/comments	None	Get comments for post.
POST /connect/posts/:id/comments	Any Auth	Add comment.
POST /connect/follow/:userId	Any Auth	Send follow request.
PATCH /connect/follow/:requestId	Target user	Accept or reject follow request.
DELETE /connect/follow/:userId	Any Auth	Unfollow.

GET /connect/messages/:userId	Any Auth	Get message thread.
POST /connect/messages/:userId	Any Auth	Send message.
GET /connect/notifications	Any Auth	Get notifications.
PATCH /connect/notifications/read-all	Any Auth	Mark all read.
GET /connect/discover	Any Auth	Get suggested users.
GET /connect/profiles/:username	None	Get public profile.
POST /connect/posts/:id/report	Any Auth	Report a post.

6.9 Admin Endpoints

Method + Endpoint	Auth	Description
GET /admin/users	Admin	List all users with filter/search.
PATCH /admin/users/:id	Admin	Update user status.
GET /admin/queue	Admin	Get all pending moderation items.
PATCH /admin/jobs/:id	Admin	Approve or remove job.
PATCH /admin/reviews/:id	Admin	Approve or reject review.
PATCH /admin/sellers/:id	Admin	Approve or reject seller application.
GET /admin/disputes	Admin	List all open order disputes.
PATCH /admin/disputes/:id	Admin	Resolve dispute. Body: { decision: refund | release, notes }.
GET /admin/revenue	Super Admin	Revenue dashboard: gross, commissions, payables, transaction list.
GET /admin/analytics	Super Admin	Platform analytics: user growth, job stats, transaction volume.
GET /admin/legal/:slug	Admin	Get legal page by slug.
PUT /admin/legal/:slug	Admin	Update legal page content.
POST /admin/email/broadcast	Admin	Send broadcast email.
GET /admin/ads	Admin	List all ads.

POST /admin/ads	Admin	Create ad.
PATCH /admin/ads/:id	Admin	Update ad.
DELETE /admin/ads/:id	Admin	Remove ad.

SECTION 7 — SHOP: FULL CATEGORY TREE

All listings must be strictly work-related, office-related, or business-related. Non-professional or unrelated items will be removed by admin. The full category tree is listed below for implementation in the database and UI.

Career Services

- CV & Applications
- Interview Preparation
- Career Coaching
- Profile Optimization
- Job Search Support
- Other Career Services (work-related only)

Professional Services

Business, Admin & Support

- Human Resources Services
- Customer Support Services
- Virtual Assistance
- Office & Administrative Support

Finance & Legal

- Bookkeeping
- Financial Advisory
- Tax Support
- Payroll Services
- Contract Drafting
- Legal Advisory

Creative & Communication

- Graphic Design
- Branding
- Video Editing
- Copywriting
- Technical Writing
- Editing & Proofreading

Health & Wellness

- Therapy / Counselling

- Workplace Wellness
- Other Health & Wellness (work-related only)

Technology & Digital

- Web Development
- Software Development
- UI/UX Design
- Data Analysis
- Cybersecurity
- IT Support
- CCTV & Security Installation

Workplace & Facility Services

- Cleaning & Maintenance
- Facility Management
- Office Setup & Installation
- Construction & Fittings
- Technical Installations
- Office Setup & Interior Services

Food & Office Meals

- Office Meals & Catering (Services)
- Meals (Bulk Orders Only)
- Snacks & Pastries (Bulk Orders Only)
- Beverages & Drinks
- Tea, Coffee & Drink Supplies
- Disposable Plates, Cups & Cutlery

Courses & Training

- Career Development
- Technology & Digital Skills
- Business & Finance
- Creative Skills
- Professional Certifications
- HSE / Safety Training

Digital Resources

Templates

- CV Templates
- Business Templates
- Design Templates

Documents

- Policies
- SOPs
- Contracts

Guides & E-books

- Career guides
- Business guides
- Industry guides

Toolkits & Packs

- HR toolkits
- Business startup packs
- Marketing kits
- Career toolkits

90-Day Plans

- Career growth plans
- Business launch plans
- Skill development plans

Office Supplies & Stationery

- Paper, notebooks, notepads
- Pens, pencils, markers
- Files, folders, binders
- Desk tools
- Desk organizers

Office Equipment & Machines

- Printers
- Scanners
- Photocopiers
- Shredders

- Laminating machines
- Binding machines
- Projectors
- Office telephones

IT, Security & Connectivity

- Routers
- Networking devices
- CCTV systems
- Biometric systems
- Computer accessories

Office Furniture

- Desks
- Chairs
- Workstations
- Cabinets
- Shelves

Interior, Décor & Office Setup

- Curtains & blinds
- Partition materials
- Glass fittings
- Wall art
- Plants
- Aquariums

Lighting & Electrical

- Lights & bulbs
- Electrical fittings
- Extension boxes

Workplace Appliances

- Air conditioners
- Fans

- Refrigerators
- Microwaves
- Water dispensers
- Coffee machines

Toiletries, Cleaning & Janitorial

- Tissue paper
- Hand wash & sanitizers
- Cleaning tools & chemicals
- Air fresheners

Safety, Medical & Emergency

- First aid kits
- Fire extinguishers
- PPE (helmets, boots, gloves, etc.)

Professional Wear & Uniforms

- Corporate wear
- Industry uniforms
- Safety wear

Business Security & Storage

- Office safes
- Lock boxes

Learning Materials

- Books
- Study guides
- Training manuals

Professional Bags & Carriers

- Laptop bags
- Briefcases

- Backpacks

Starter Kits & Bundles

- Job seeker kits
- Office setup kits
- Remote work kits

Industrial, Oil & Gas, Drilling & Mud Engineering

- PPE & safety gear
- Drilling chemicals
- Mud engineering tools
- Solids control equipment
- Industrial measuring tools

Other / Uncategorized

For listings that do not fit any existing category. All listings must be strictly work-related, office-related, or business-related. Non-professional or unrelated items will be removed.

- Miscellaneous Services (work-related only)
- Miscellaneous Products (work/office-related only)

Requires seller checkbox confirmation: 'I confirm this listing is work-related or office-related'

SECTION 8 — SECURITY REQUIREMENTS

8.1 Authentication & Authorisation

Requirement	Implementation
Password hashing	bcrypt with 12 salt rounds. Never store plaintext passwords.
JWT access token	RS256 or HS256. 15-minute expiry. Payload: { user_id, role, email }
JWT refresh token	7-day expiry. httpOnly, Secure, SameSite=Strict cookie. Never accessible via JavaScript.
Token rotation	New refresh token on every /auth/refresh. Old token immediately invalidated in Redis.
Role-based access control	NestJS Guards enforce role requirements on every protected route server-side.
Email verification	Required before accessing dashboard features. Token expires 24 hours.
Age verification	Date of birth validated on registration: must be 18+. Stored for legal compliance.
Terms of Service consent	tos_agreed_at timestamp stored in DB at moment of registration.
Account deletion	Soft delete: status = deleted, personal data anonymised, all active sessions invalidated.

8.2 Spam & Abuse Protection

Area	Protection Measure
Registration	Google reCAPTCHA v3 on sign-up. Score threshold: 0.5.
Anonymous review/salary	reCAPTCHA v3 + hashed IP+UA fingerprint for duplicate detection (internal only, never exposed).
API rate limiting	Redis-backed via Nginx: 100 req/min per IP global. Auth: 5 req/min. Review submit: 3 per IP/hour.
Job applications	UNIQUE constraint on (seeker_id, job_id). Enforced at API and DB levels.
Shop orders	HMAC-SHA512 webhook signature verified before any order status update.
Escrow abuse	Dispute window limited to 3–6 days. Repeat fraudulent disputes trigger admin review of buyer account.
File uploads	MIME type validated server-side. File size limits enforced. Phase 2: ClamAV antivirus scan.

Content moderation	All jobs, reviews, and seller applications require admin approval before going live.

8.3 Data Security

Area	Measure
Transport security	HTTPS enforced everywhere via Cloudflare SSL. HSTS header set.
SQL injection	TypeORM parameterised queries. No raw SQL string concatenation.
XSS prevention	Helmet.js sets CSP, X-Content-Type-Options, X-Frame-Options. Input sanitised before DB write.
CORS	Whitelist only the known frontend origin (Vercel URL). Reject all other origins.
S3 file security	Private bucket. No public URLs. All downloads via pre-signed URLs (1-hour expiry). S3 keys never returned in API responses.
Flutterwave webhook	HMAC-SHA512 verified using Flutterwave secret key before processing. Invalid signatures return 401.
Salary data anonymity	user_id never stored on salary_reviews. Hashed fingerprint for spam detection only — never surfaced in API.
Review anonymity	display_name (nickname) shown only. Real name never returned by API even to admins.
Environment variables	All secrets in environment variables, never in source code. AWS Secrets Manager in production.
Input validation	class-validator DTOs in NestJS validate all request bodies before service layer.

8.4 Infrastructure Security

Requirement	Implementation
DDoS protection	Cloudflare Pro in front of all infrastructure.
Database access	PostgreSQL accessible only from API server private IP. No public endpoint. VPC/security groups on AWS.
Server hardening	SSH key-based access only. Fail2ban on SSH. Minimal open ports: 80, 443, 22 (restricted IP).
Backups	Automated daily PostgreSQL dump to S3. 30-day retention. Weekly integrity test.
Logging	All API requests logged (method, path, response code, latency). No PII in logs. Sentry captures all 5xx.

Monitoring	UptimeRobot pings key endpoints every 5 minutes. PagerDuty alert on downtime > 2 minutes.
Dependencies	npm audit in CI/CD. Dependabot alerts for security vulnerabilities.

8.5 SEO & Performance

Requirement	Implementation
Server-side rendering	Next.js SSR on Jobs, Reviews, Salaries, Shop, and all legal pages for full Google crawlability.
Dynamic meta tags	og:title and og:description per job post, company page, and shop product. Canonical URLs.
Sitemap	Auto-generated sitemap.xml with all active jobs, company review pages, and shop products.
Core Web Vitals	LCP < 2.5s · CLS < 0.1 · FID < 100ms — monitored via Vercel Analytics.
Image optimisation	Next.js Image component (WebP, lazy loading, responsive). Cloudinary for seller thumbnails.
API response caching	Redis TTL: GET /jobs (5min) · GET /reviews/companies (10min) · GET /salaries aggregates (30min).
robots.txt	Disallow: /admin/, /api/, /dashboard/. Allow all public pages.

8.6 Phase 1 MVP Launch Checklist

#	Item
1	User registration (seeker + employer) with email verification working end-to-end
2	Employer job posting → admin approval → live on jobs page flow complete
3	Job search with all 10 filters functional
4	Company review submission (guest + user) → moderation → publish flow complete
5	Salary submission + aggregated display working
6	Shop: product/service listing upload, cart, Flutterwave checkout, escrow, download link functional
7	Shop: Quote request → seller responds → buyer accepts → payment → escrow flow functional
8	Shop: Buyer confirmation window + auto-release cron job tested
9	Shop: Dispute flow → admin resolution tested

10	Admin dashboard: all moderation queues working including disputes
11	All 12 legal pages published and accessible from footer
12	Legal pages editable in admin WYSIWYG editor
13	Connect: post, like, comment, follow, DM functional
14	Password reset flow end-to-end tested
15	Rate limiting and CAPTCHA active on all public forms
16	HTTPS enforced, security headers verified
17	Flutterwave webhook HMAC verification tested with live credentials
18	Database backup system active and tested
19	Mobile responsive across all screen types
20	Sentry error monitoring active
21	Privacy Policy and Terms of Service reviewed by legal counsel
22	Marketplace Policy and Refund Policy reviewed and published